BBST Magazine -- Volume 02 / 2025

Official Student Publication - Faculty of Technological Studies, UWU

Welcome to the second annual volume of BBST Student Magazine.

This issue highlights advances in agricultural automation, climate-smart robotics,

and student entrepreneurial breakthroughs across Uva province.

Explore cutting-edge research, faculty perspectives, and artistic profiles.

Cover Story: Autonomous Agricultural Rovers engineered in Passara Road Campus.

Published by Uva Wellassa University of Sri Lanka, Badulla.

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Dean’s Welcome Message

Prof. H. M. Wickramasinghe -- Dean, Faculty of Technological Studies

It is an immense privilege to present the second edition of BBST Magazine.

Our faculty was founded on the vision of producing forward-thinking technologists

who blend scientific rigor with practical industrial problem-solving.

The student contributions in this issue reflect our focus on sustainable dry-zone agriculture,

automated food processing systems, and smart bio-industrial technology.

I congratulate the editorial committee and every student author who has contributed.

Autonomous Robotics in Tropical Agriculture

Lead Research Paper -- Kasun Perera & Dr. T. Senanayake

Precision farming in Sri Lanka faces distinctive micro-climatic challenges.

This paper reports on our experimental solar-assisted unmanned rovers designed

to monitor paddy fields in Monaragala and Polonnaruwa.

Using lightweight multispectral sensors and computer vision algorithms, the rover

detects fungal blight 72 hours before visible leaf discoloration occurs.

Field trials demonstrated a 34 percent reduction in chemical pesticide usage.

Plant Nanotechnology & Crop Resilience

Applied Biosystems Review -- Dinithi Rajapaksha & Dr. R. Silva

Nanoscale delivery vehicles are transforming conventional agrochemical distribution. By encapsulating zinc and potassium nanoparticles within biodegradable chitosan shells, we achieved sustained nutrient release across a 28-day growth cycle. Maize crops treated with nano-formulations showed 22 percent greater root depth and significant drought resistance under elevated ambient temperatures.

Student Innovation Spotlight: AgroTech Expo

Highlighting Top Prototype Projects from the Class of 2025

1. Automated Cinnamon Peeling Robotic Arm - First Prize Winner

2. IoT Cold-Chain Monitoring System for Perishable Spices

3. Microbial Fuel Cells Utilizing Tea Factory Effluent

4. Drone-Mounted Thermal Imaging for Wild Elephant Herd Early Warning

Each project was evaluated by a joint academic-industry jury from MAS, Hayleys, and SLT.

Department Highlights & Industry Partners

Building Pathways between University Research and National Enterprise

During the past year, the Biosystems Technology Department secured over LKR 18M in competitive research grants from the National Science Foundation (NSF).

Four patent applications were filed by undergraduate research teams.

We express gratitude to our industry mentors who provide student internships, field trial access, and commercialization pathways.

BBST Magazine -- Back Cover

Student Voices for a Brighter Tomorrow

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Call for Papers Volume 03 / 2026 opens August 2025.

Thank you for reading BBST Student Magazine.